

MARIA C.



PUNTUACIÓN MÁS ALTA 49K



KEVIN L.

PROYECTO ADOA Y PCO

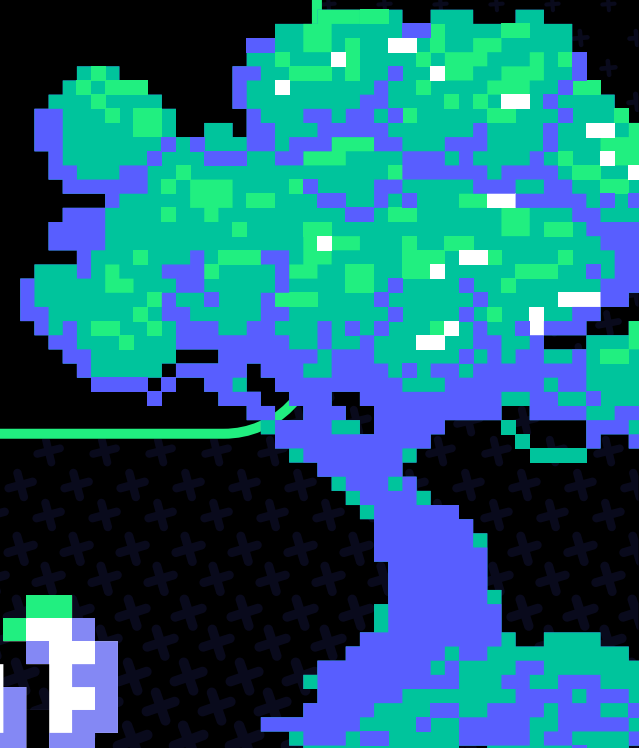
START

MENU

SIGN IN



◆ MARIA JOSE CASTAÑEDA GRAJALES
◆ KEVIN SANTIAGO LOPEZ SALAZAR



MENU

01

02

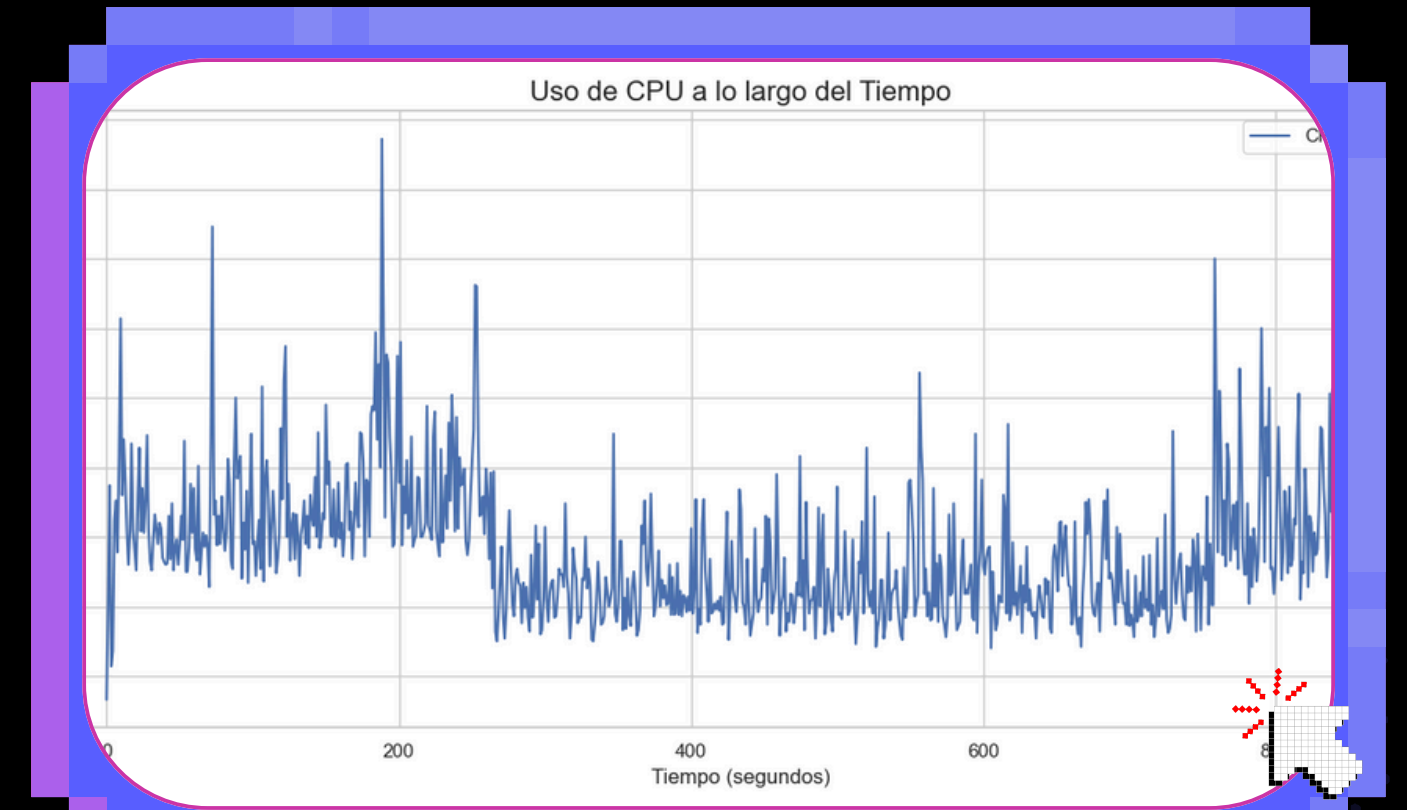
03



GEOMETRIC

➡ ALGORITMO SECUENCIAL. CONSTA DE 3 ETAPAS:

1. CALCULAR TABLA DE COSTOS USANDO PROGRAMACION DINAMICA
2. IDENTIFICAR CANDIDATOS FILTRANDO PARTICIONES TRIVIALES
3. EVALUAR CANDIDATOS FINALES



PIVOT

VOLVER A LA PÁGINA
DEL CONTENIDO

MENU

02

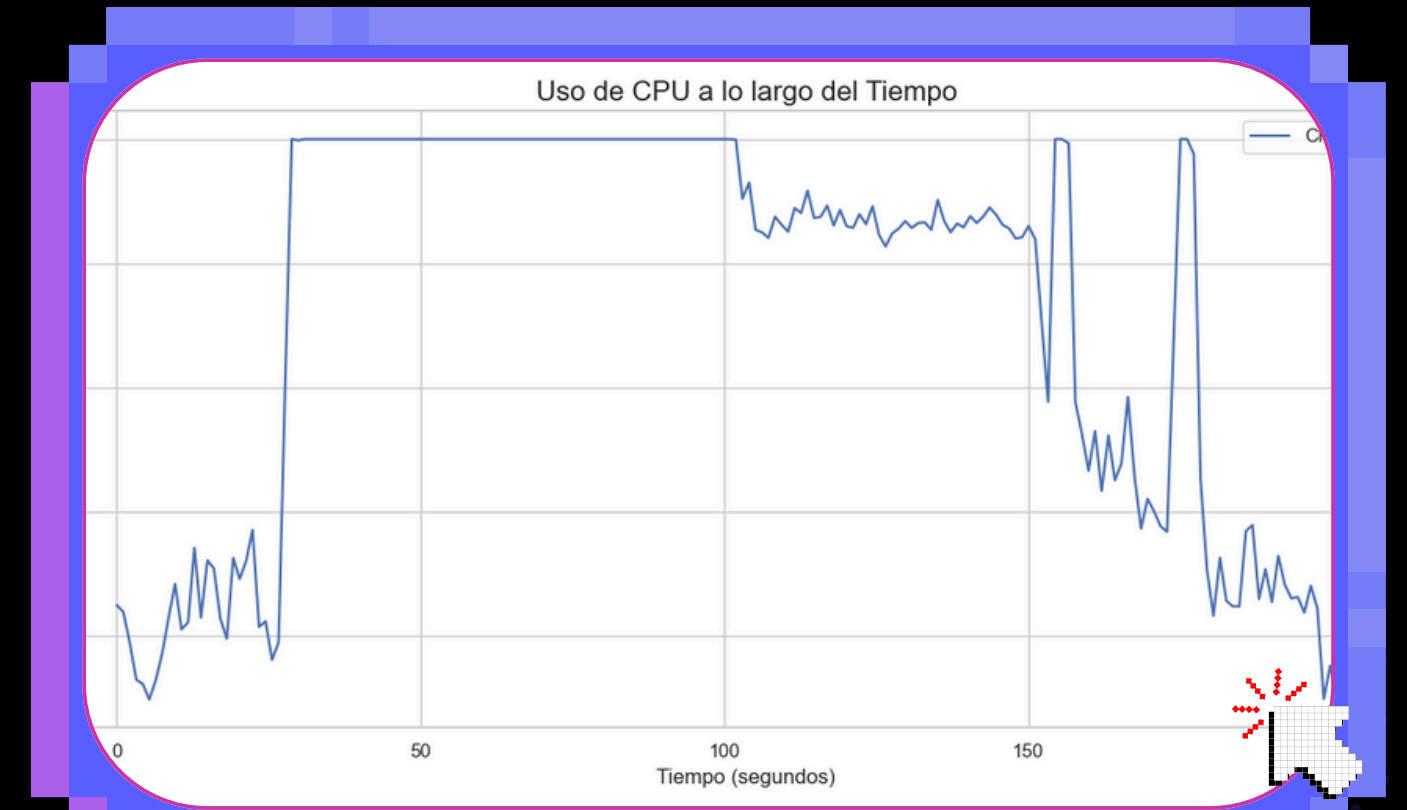
04

06



GEOMETRIC MP

- ➡ ALGORITMO PARALELIZADO CON MULTIPROCESSING. CONSTA DE 3 ETAPAS:
1. CALCULAR TABLA DE COSTOS USANDO PROGRAMACION DINAMICA Y MP
 2. IDENTIFICAR CANDIDATOS USANDO MP Y FILTRADO DE PARTICIONES TRIVIALES
 3. EVALUAR CANDIDATOS FINALES USANDO MP



MP

VOLVER A LA PÁGINA
DEL CONTENIDO

MENU

10

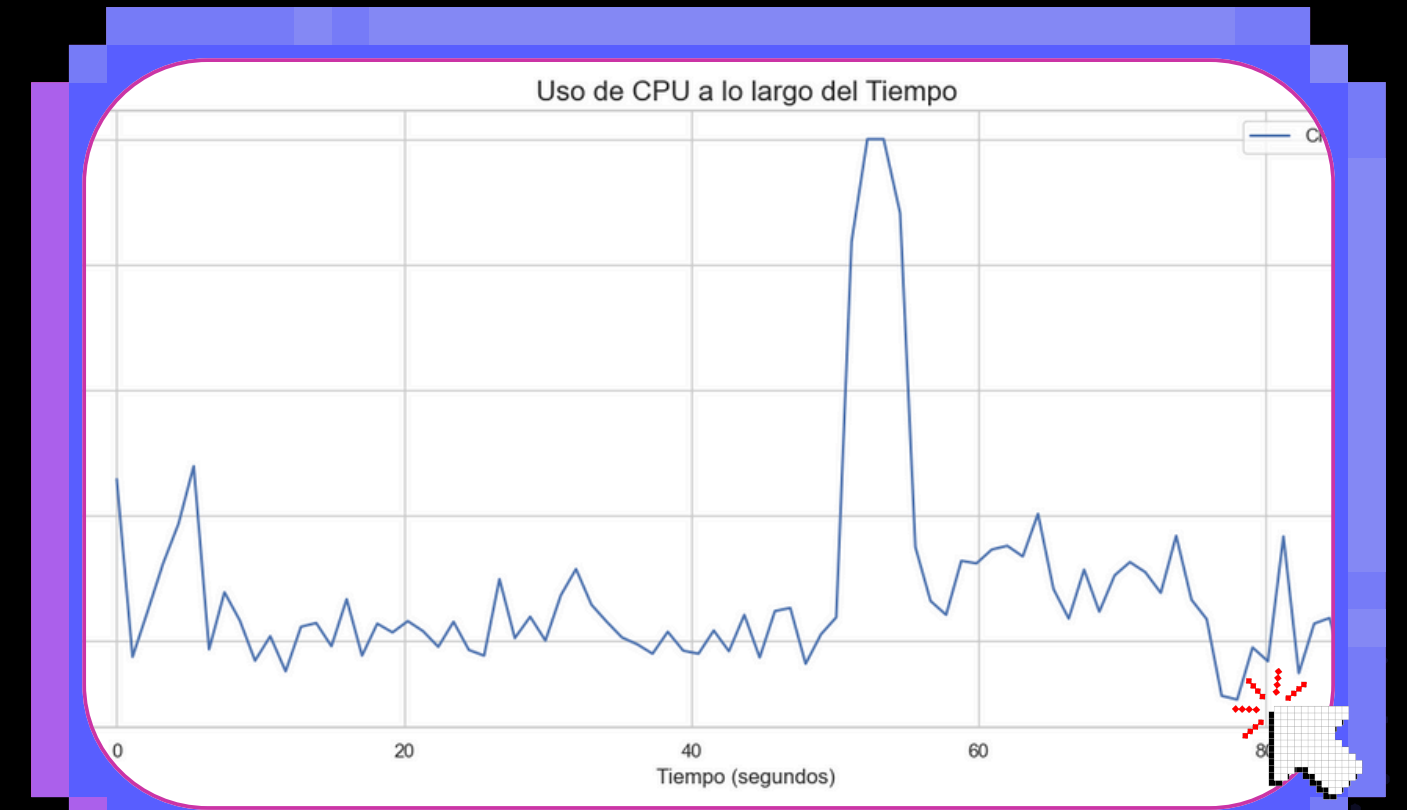
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10



GEOMETRIC MP+CUOA

- ➡ ALGORITMO PARALELIZADO CON CUDA Y MULTIPROCESSING. CONSTA DE 3 ETAPAS:
1. CALCULAR TABLA DE COSTOS USANDO PROGRAMACION DINAMICA Y CUDA
 2. IDENTIFICAR CANDIDATOS FILTRANDO PARTICIONES TRIVIALES
 3. EVALUAR CANDIDATOS FINALES PARALELIZADO CON MULTIPROCESSING



MP+CUOA

VOLVER A LA PÁGINA
DEL CONTENIDO

JUGADOR 1

PRUEBAS



[VOLVER A LA PÁGINA
DEL CONTENIDO](#)

[illegible][illegible]

Proyecto (Pro)				Proyecto Paralelo MP				Proyecto Paralelo MP			
	Perdida Q	Tiempo Q	Particion	Perdida Pro	Tiempo Pro	Particion	Perdida MP	Tiempo MP	Particion		
OP,Q,R,S,T	0.50	1083.60	J A,B,C,D,E,F,G,H,I,K,L,M,N,O,P,Q,R,S,T	0.50	413.19	J A,B,C,D,E,F,G,H,I,K,L,M,N,O,P,Q,R,S,T	0.50	88.44	J A,B,C,D,E,F,G,H,I,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	132.17	B A,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	183.79	M A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	45.84	M A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	256.90	L A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	185.45	L A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	46.55	L A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	139.47	M A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	87.71	M A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	27.98	M A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	4.84	G A,B,C,D,E,F,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	0.23	G A,B,C,D,E,F,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	11.19	G A,B,C,D,E,F,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	3.98	R A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	0.21	R A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	11.15	R A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	10.13	R A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	4.58	R A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	12.00	R A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.50	917.21	J A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.50	369.40	J A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.50	81.78	J A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	186.09	B A,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	187.88	M A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	43.90	M A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	259.46	L A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	196.76	L A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	44.22	L A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	128.31	M A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	88.09	M A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	26.84	M A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	3.91	G A,B,C,D,E,F,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	0.20	G A,B,C,D,E,F,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	10.97	G A,B,C,D,E,F,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	4.03	R A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	0.20	R A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	10.93	R A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	9.38	R A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	5.00	R A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	12.01	R A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.50	1006.57	J B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.50	382.28	J B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.50	87.78	J B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	181.90	M B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	172.86	M B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	45.03	M B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	256.27	L B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	179.64	L B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	46.98	L B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	131.72	M B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	88.62	M B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	27.63	M B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	G Titulo del gráfico		G B,C,D,E,F,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	0.21	G B,C,D,E,F,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	11.85	G B,C,D,E,F,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	4.19	R B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	0.21	R B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	11.49	R B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	10.35	R B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	4.40	R B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	13.97	R B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.50	759.19	J B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.50	357.04	J B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.50	85.94	J B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	160.50	M B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	165.32	M B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	45.53	M B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	165.16	L B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	167.27	L B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	47.75	L B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	57.56	M B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	77.71	M B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	28.47	M B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	2.95	G B,C,D,E,F,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	0.20	G B,C,D,E,F,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	12.08	G B,C,D,E,F,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	2.90	R B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	0.19	R B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	11.31	R B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.00	6.51	R B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	3.78	R B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T	0.00	12.40	R B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T		
OP,Q,R,S,T	0.50	304.00	K A,C,E,G,I,M,O,Q,S	0.50	184.00	K A,C,E,G,I,M,O,Q,S	0.50	45.14	K A,C,E,G,I,M,O,Q,S		
OP,Q,R,S,T	0.00	56.84	M A,C,E,G,I,K,O,Q,S	0.00	92.99	M A,C,E,G,I,K,O,Q,S	0.00	26.94	M A,C,E,G,I,K,O,Q,S		

➡ PRUEBAS REALIZADAS DESDE 3 A 21 VARIABLES



RESUMEN

	GEOMETRIC			GEOMETRIC + Multiprocessing			GEOMETRIC + Multiprocessing + CUDA		
Tamaño	MIN	PROM	MAX	MIN	PROM	MAX	MIN	PROM	MAX
3	0.00%	7.65%	25.00%	0.00%	7.65%	25.00%			
5	0.00%	1.80%	50.00%	0.00%	1.80%	50.00%			
10	0.00%	0.04%	2.08%	0.00%	0.04%	2.08%			
15	0.00%	1.00%	3.00%	0.00%	1.00%	3.00%	0.00%	1.00%	3.00%
20	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
21									
Promedio	0.00%	2.10%	16.02%	0.00%	2.10%	16.02%	0.00%	0.50%	1.50%

PROMEDIO SPEEDUP									
	GEOMETRIC			GEOMETRIC + Multiprocessing			GEOMETRIC + Multiprocessing + CUDA		
Tamaño	MIN	PROM	MAX	MIN	PROM	MAX	MIN	PROM	MAX
3	0.01	0.49	1	-15.94	-11.46	-10.66			
5	-0.01	0.33	1	-1475	-170.13	-9.47			
10	0.12	298.79	1408	-1371	-495.7	-12.3			
15	-7.09	120.13	1940	-1161	-152.6	-1.88	-438	-57.75	1.14
20	-1.86	4.76	21.04	-112.8	-19.53	4.67	-44.45	-0.25	18.72
21	0	0	0	-85.82	-9.25	6.01	-36.82	7.85	46.86
Promedio	-1.47	70.75	561.84	-703.59	-143.11	-3.94	-173.09	-16.72	22.24

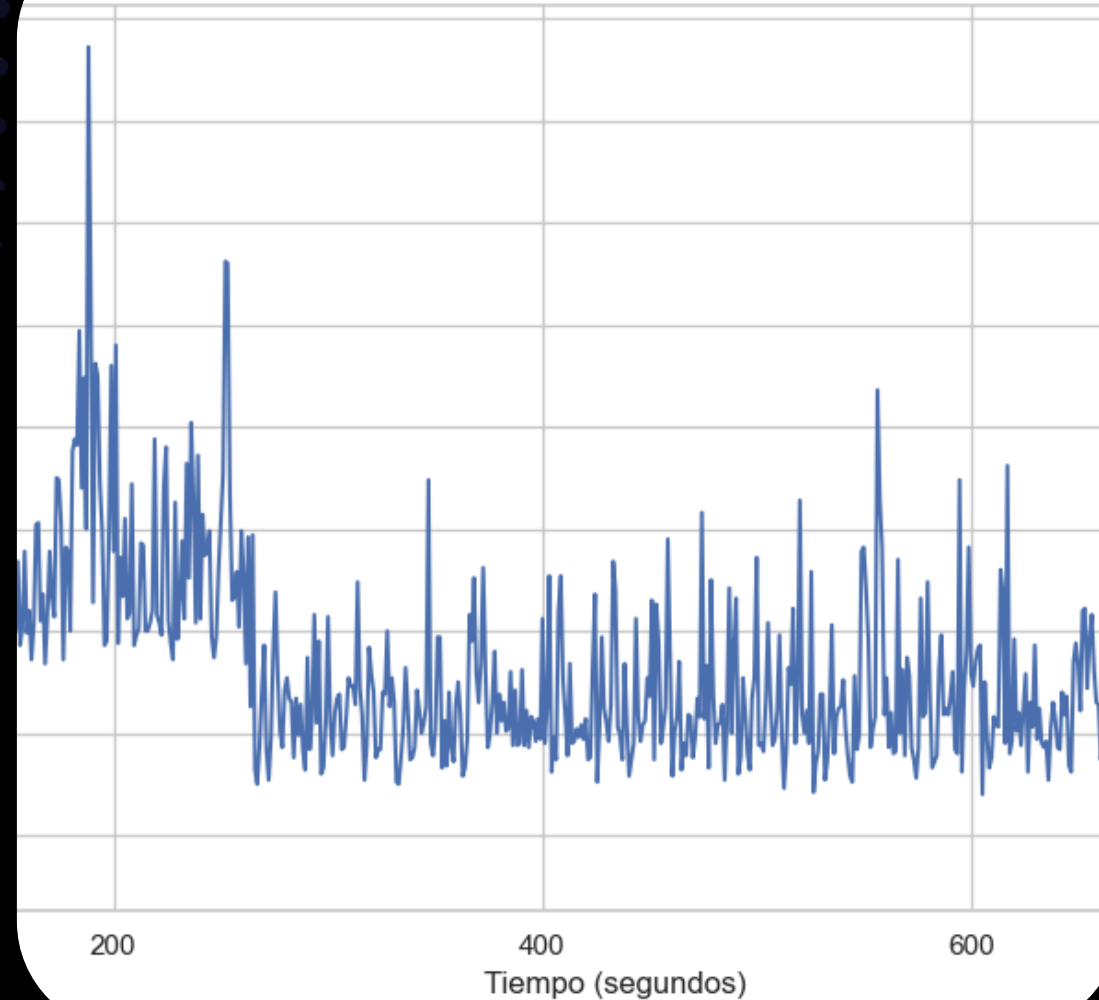


CONSUMO GEOMETRIC

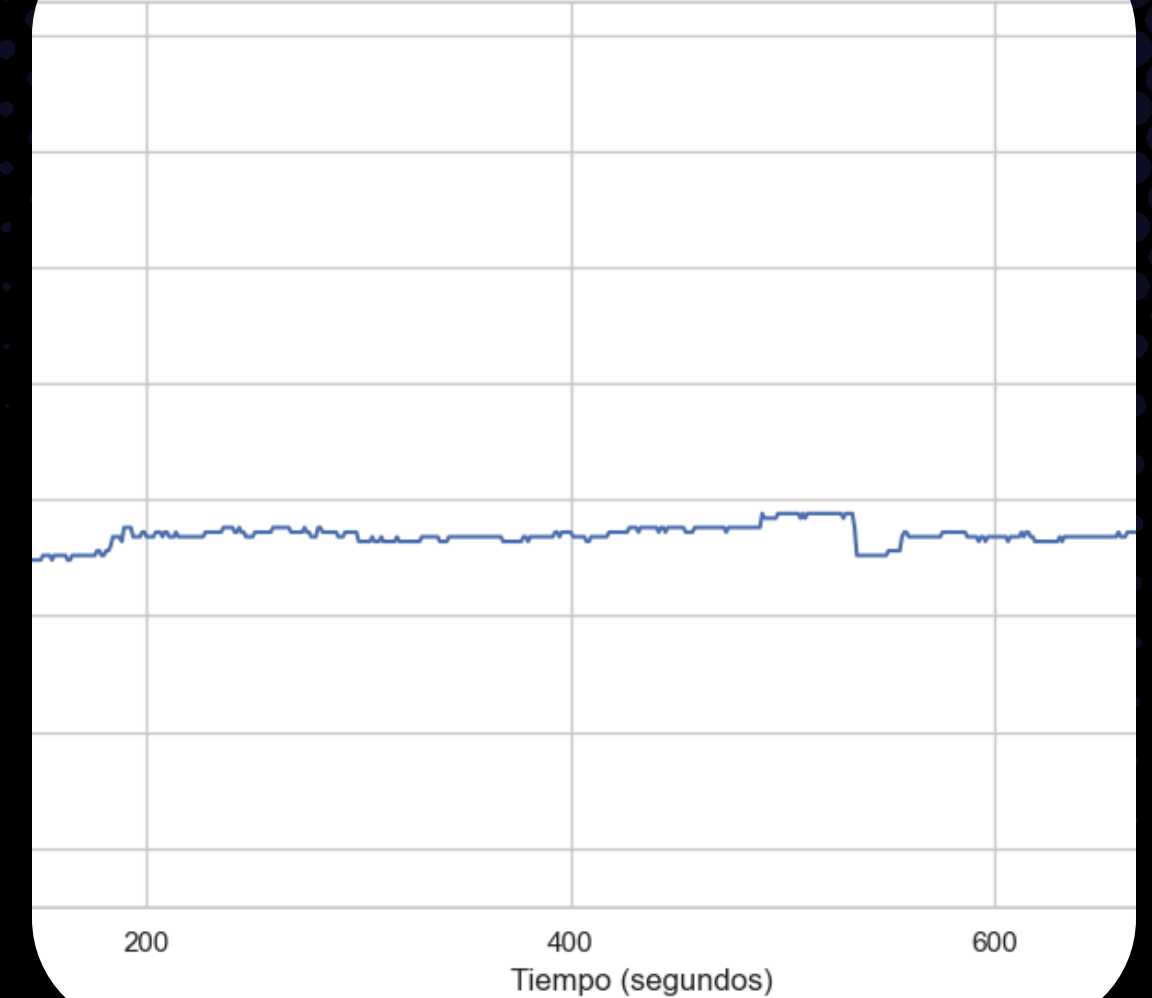


VOLVER A LA PÁGINA
DEL CONTENIDO

Uso de CPU a lo largo del Tiempo



Uso de Memory a lo largo del Tiempo



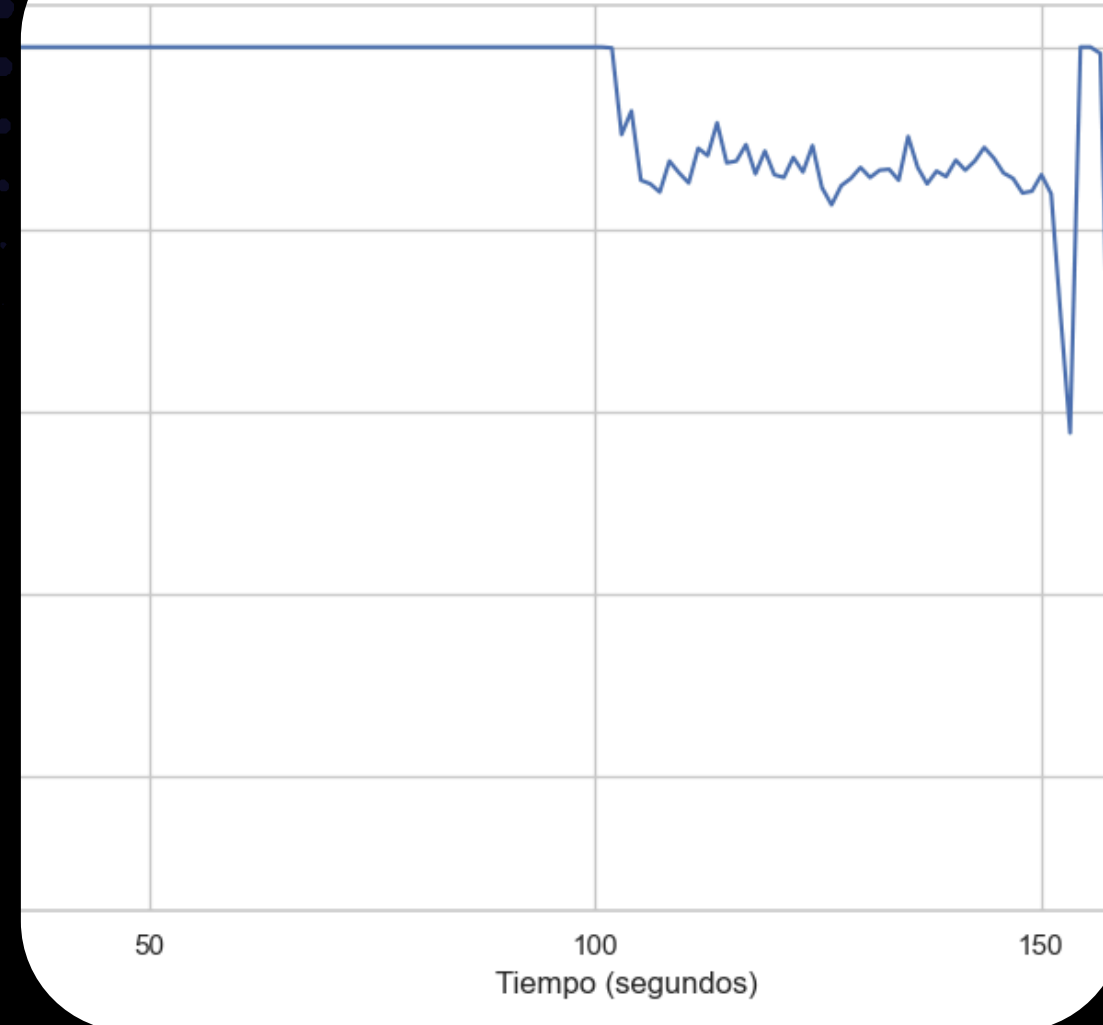
➡ MUCHO MAS DEMORADO, PERO POCO USO DE MEMORIA Y CPU

CONSUMO GEOMETRIC MP

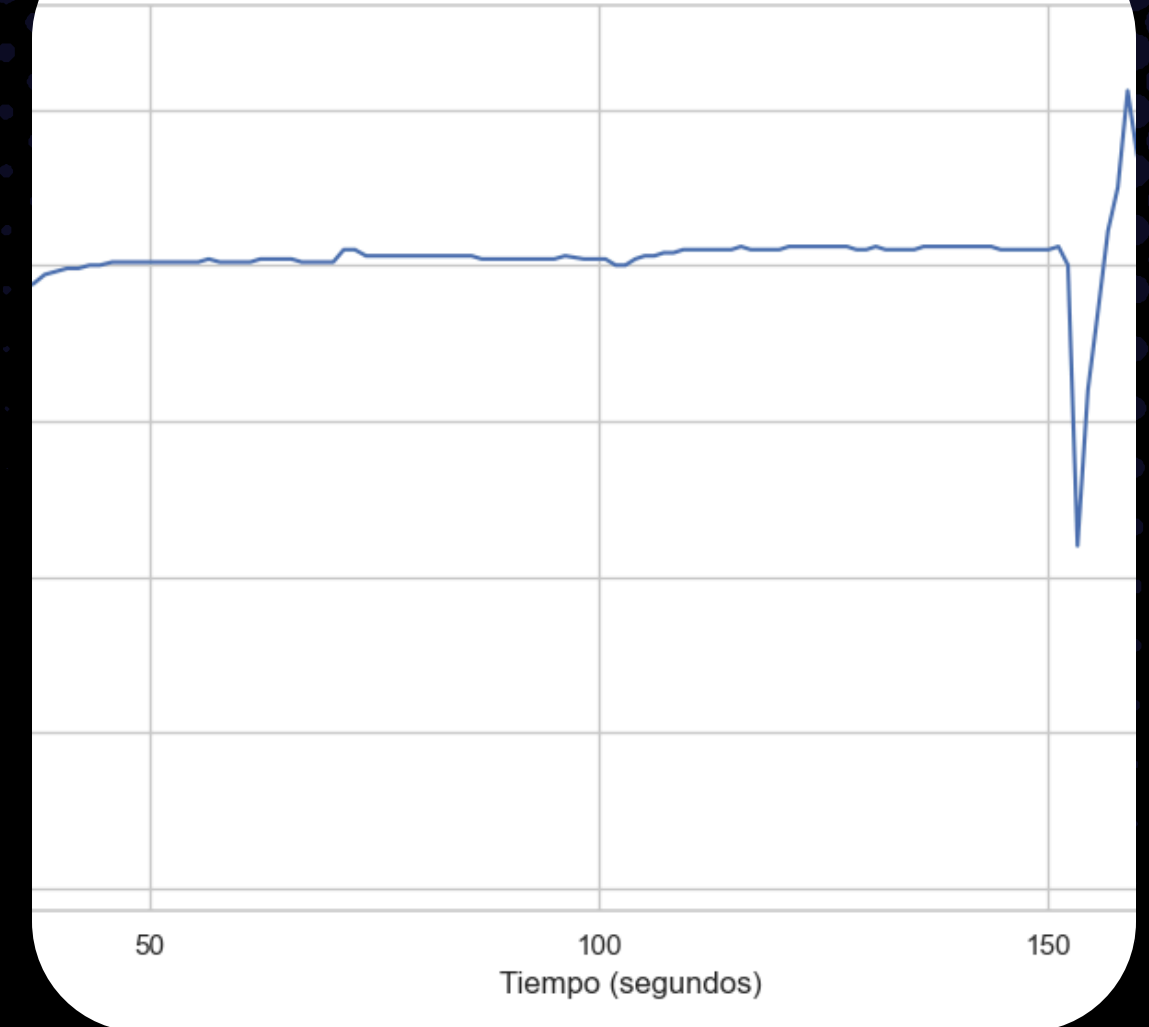


VOLVER A LA PÁGINA
DEL CONTENIDO

Uso de CPU a lo largo del Tiempo



Uso de Memory a lo largo del Tiempo

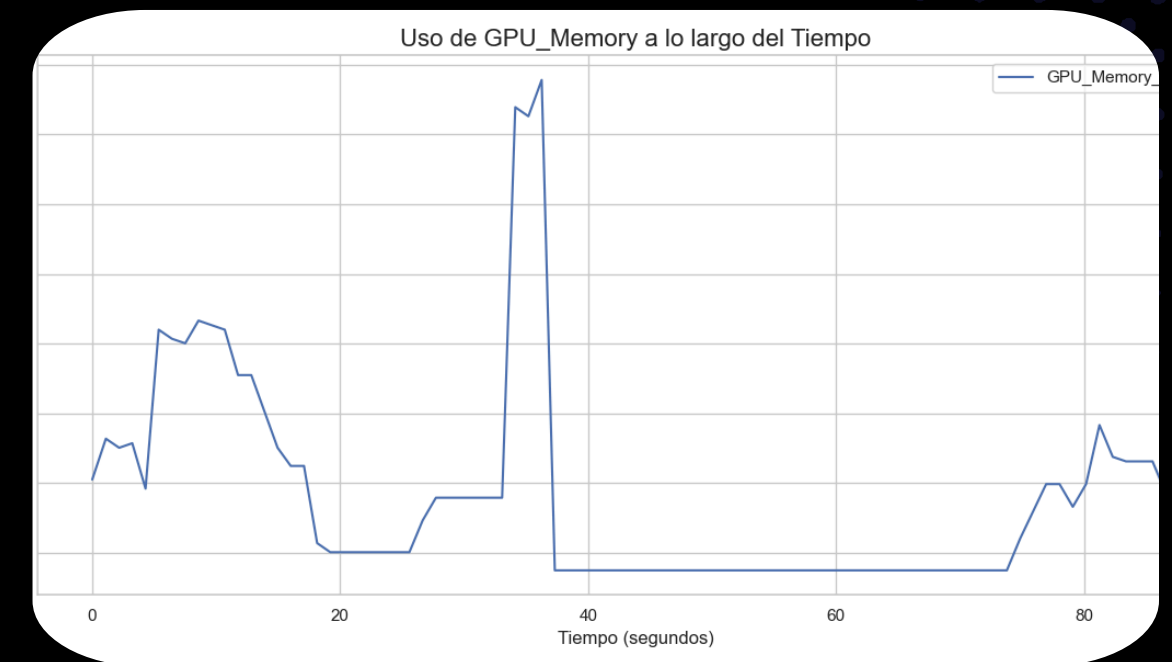
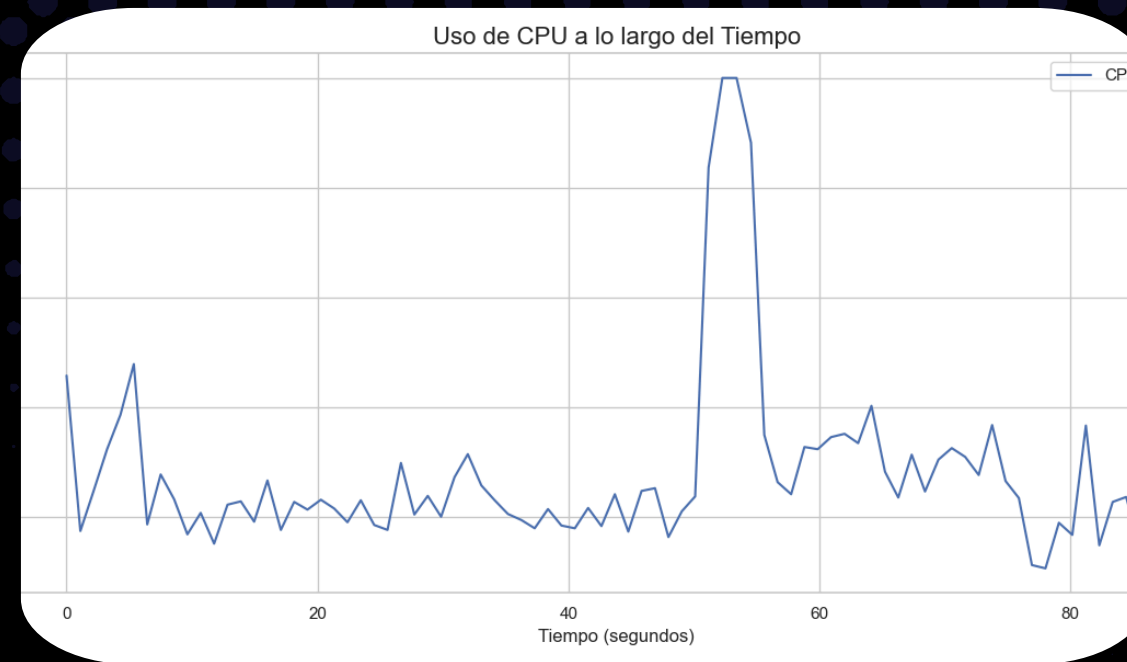


➡ MAS EFICIENTE, PERO CONSUMO DE CPU AL 100%
[CONFIGURABLE]. LA MEMORIA TAMBIÉN AUMENTA
CONSIDERABLEMENTE.

CONSUMO GEOMETRIC MP + CUDA



VOLVER A LA PÁGINA
DEL CONTENIDO



➡ MUCHO MAS EFICIENTE, SE ALCANZA EL 100% DE CPU, USO DE MEMORIA MODERADO, USO DE GPU SUBUTILIZADA (PERO MUCHO MAS EFICIENTE QUE MP)

MENU

01

07

12



COMPARATIVA TIEMPOS (20N)

✦ PARA LA PRUEBA MAS PESADA
[NORMALMENTE LA PRIMERA] ARROJA LOS
MEJORES RESULTADOS

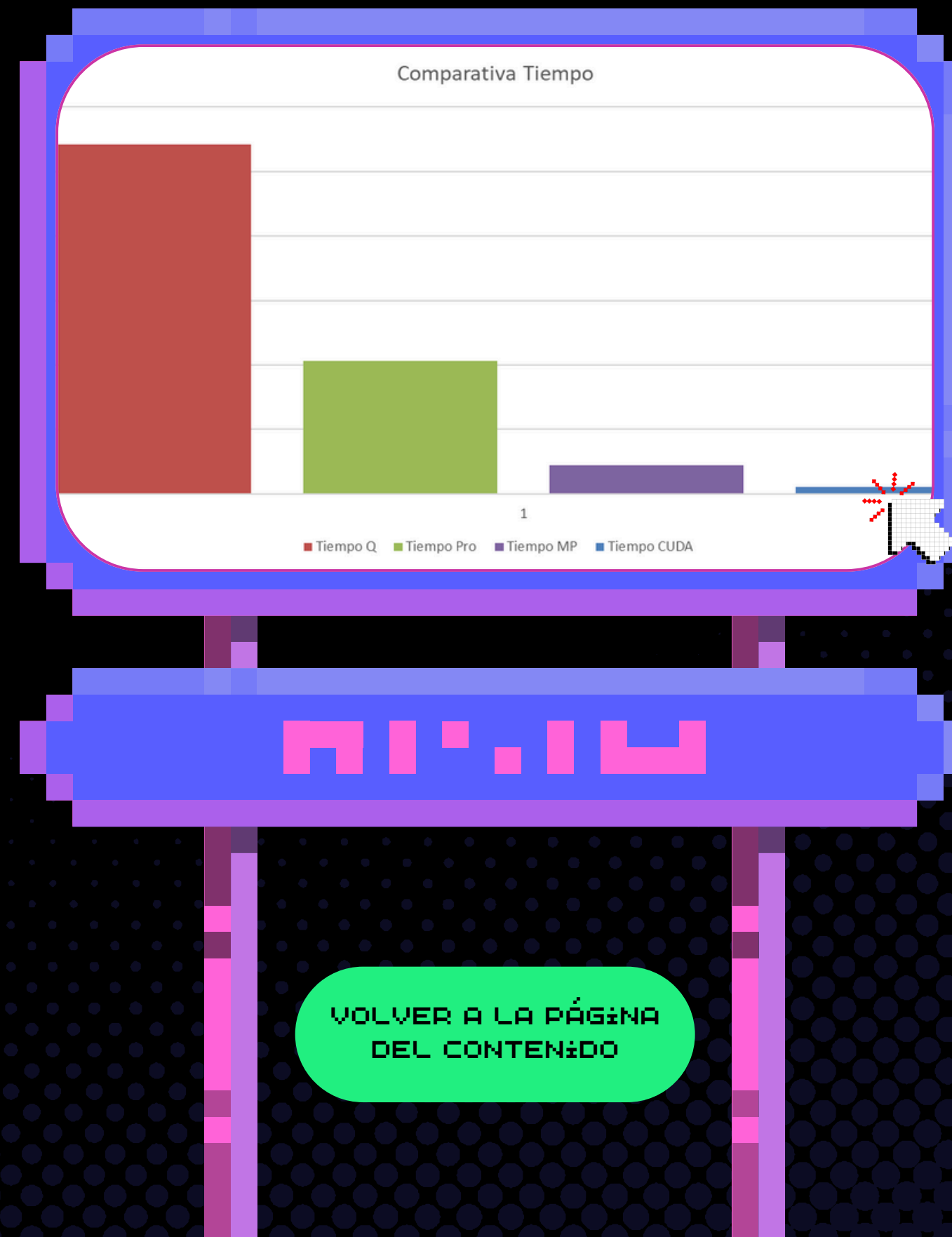
QNODES → 1083 S

GEOMETRIC → 413 S

GEOMETRIC MP → 88 S

GEOMETRIC MP + CUDA → 22 S

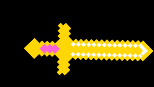
SPEEDUP FINAL = 49X



PIV

VOLVER A LA PÁGINA
DEL CONTENIDO

MENU



01



07



12



COMPARATIVA TIEMPOS (21N)



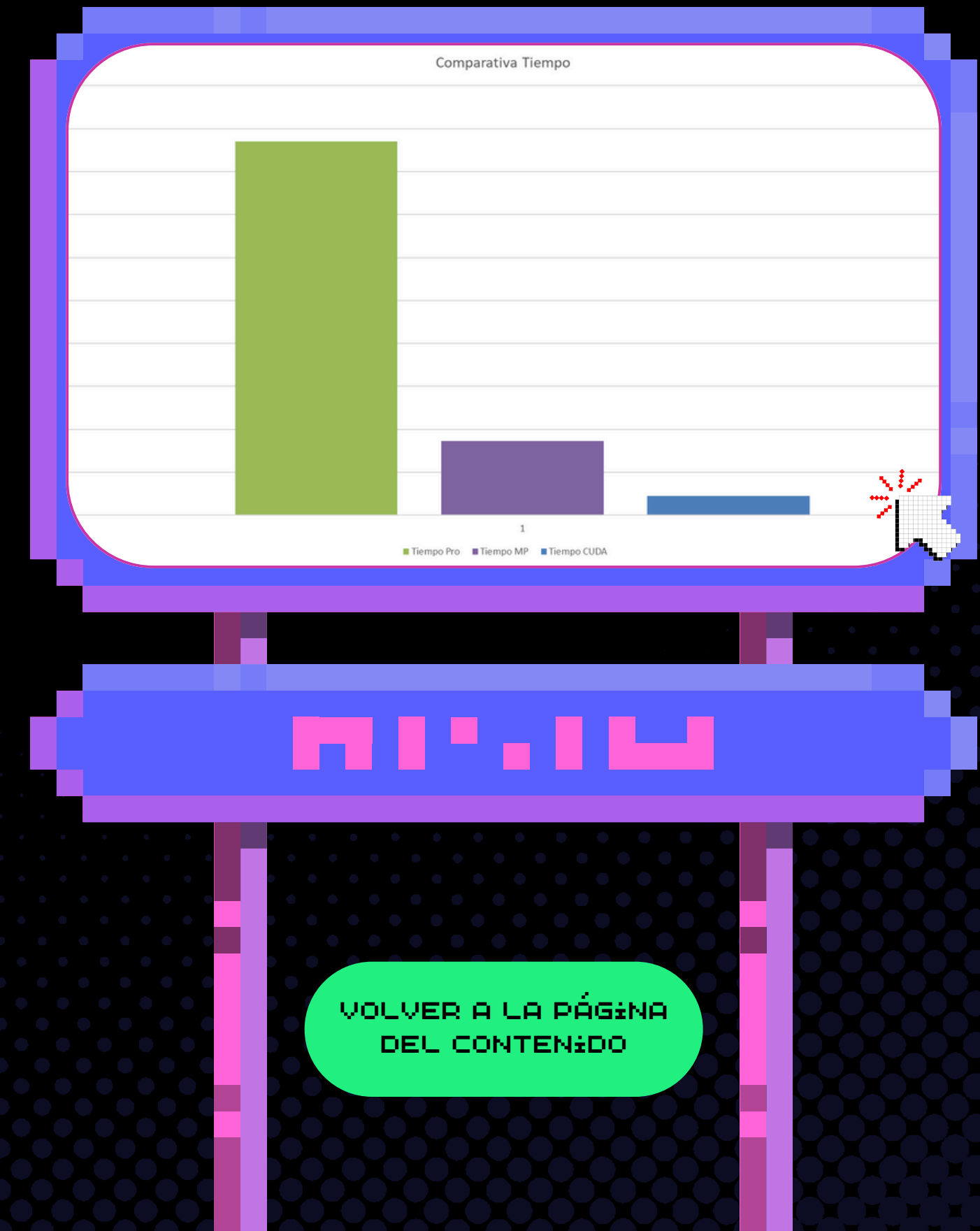
PARA LA PRUEBA MAS PESADA
[NORMALMENTE LA PRIMERA] ARROJA LOS
MEJORES RESULTADOS

GEOMETRIC → 870 S

GEOMETRIC MP → 173 S

GEOMETRIC MP + CUDA → 45 S

SPEEDUP FINAL = 19X

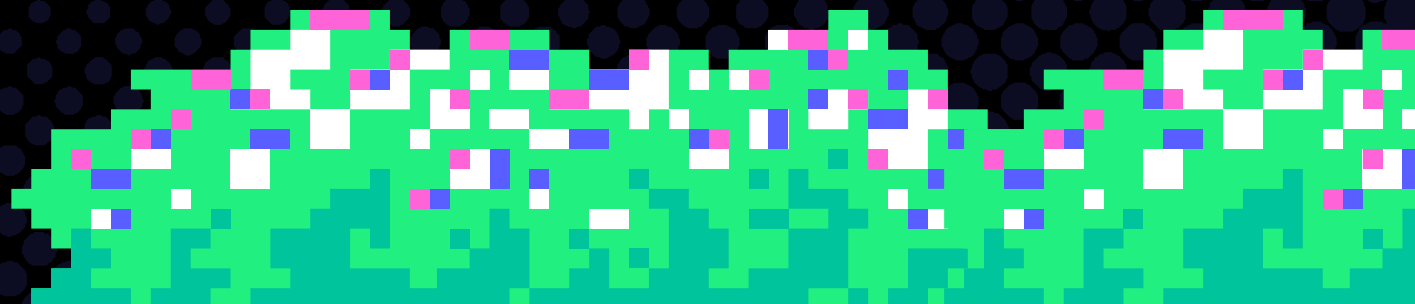
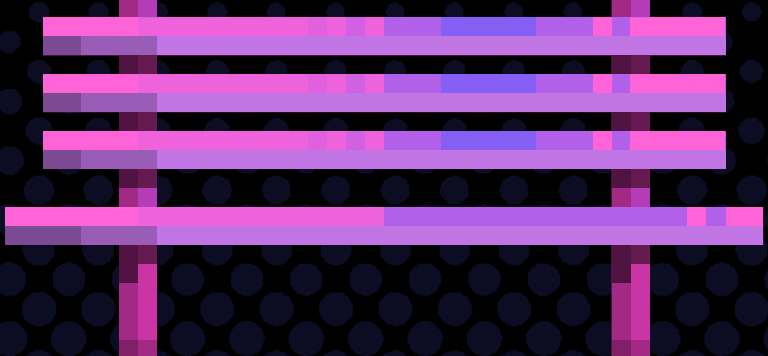
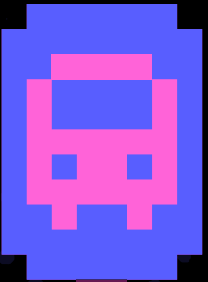


PIV

VOLVER A LA PÁGINA
DEL CONTENIDO



MENU



COMPARACION TIEMPOS POR FUNCION EN 21N					
Proceso	Geometric (s)	Geometric MP (s)	Geometric MP CUDA (s)	SpeedUp MP (X)	SpeedUp CUDA (X)
1	0.35	0.33	0.27	1.06	1.3
2	829.35	148.22	4.28	5.6	193.77
3	14.05	20.03	13.32	0.7	1.05
4	0.23	22.95	23.72	0.01	0.01
5	843.98	191.53	41.59	4.41	20.29



1	Descomposicion en Tensores
2	Calculo de la tabla de costos
3	Identificacion de Candidatos + EXTRA + Filtrado
4	Evaluacion de candidatos
5	Tiempo Final

SIGN IN



VOLVER A LA PÁGINA DEL CONTENIDO



FIN

CONSULTAR PDF Y EXCEL PARA MAS DETALLES

